

Queueing-Based Rate Allocation and Routing in Small Networks

Aravind Muraleedharan, Javier Pérez Calderón, Shrey Parikh

School of Electrical and Computer Engineering, Purdue University

West Lafayette, IN, USA

Email: amuralee@purdue.edu, parikh94@purdue.edu, perezcal@purdue.edu

Abstract—Classical queueing models provide theoretical insight into stability, delay, and loss in packet-switched systems. This project studies practical simulations of these models in a small network setting. We first verify canonical single-queue results using M/M/1, D/D/1, and finite-buffer M/M/1/N by comparing simulation estimates of key metrics (e.g., mean delay and mean backlog) against known theoretical expressions. We then simulate a multi-hop routed testbed with finite buffers and service-rate constraints at each link, and evaluate how different routing policies affect throughput, delay, and drop probability under increasing load. Finally, we compare our queue-aware routing against a static shortest-hop routing.

Index Terms—packet networks, routing, rate allocation, multi-hop networks, finite buffers

I. INTRODUCTION

Server to server communication relies mainly on packet switching, wherein the information sent is divided between many packets which follow their own route to the destination. Furthermore, these packets rely on an existing network infrastructure based on routers and servers to navigate to their destination. This network ensemble relies on the simultaneous implementation of numerous systems, including routing, congestion control, header processing, data processing, queue control, encoding, data encryption, etc. In our present work, we will focus on queue control, and how the various models on queue behavior function.

More concretely, with this work we set out to verify whether three types of queues following Kendall's notation have the theoretical steady-state number of packets, average number of packets, and processing time per packet that their equations predict. This demonstration serves to reinforce our confidence in present queueing infrastructure. In addition, we have graphed the time evolution of the number of packets and the processing time between packets before reaching the steady state.

Following our study, we have found that: (i) the canonical single-queue models provide accurate predictions for the mean delay and mean backlog when their assumptions are matched in simulation, and (ii) in small networks, queueing-informed routing/rate allocation can reduce congestion at critical links and improve end-to-end delay and loss under high loads.

II. RELATED WORK

The theoretical and mathematical behavior of classical queueing models has been studied extensively; standard refer-

ences include Kleinrock [1], Harchol-Balter [2], and lecture notes/slides such as Jain [3] and Güneş [4]. The present investigation has relied primarily on the course material for the Purdue ECE 54700 course as a base on which to build the relevant queue simulations. This work is not intended to be a review of the existing literature on queueing theory, but rather an exploration of how said theory holds up in real world case studies. Our small-network motivation relates to congestion management and routing policies that react to congestion. Wu *et al.* [5] discuss a control-theoretic approach to in-network congestion management. Backpressure routing has also been evaluated in practical settings (e.g., via emulation/testbeds) [6]. Rate-shaping mechanisms such as token bucket and hop-by-hop rate control have been used to manage QoS and multimedia/network performance [7]–[9]. Our work is intentionally lightweight, and we use queueing expressions as a design heuristic and validate the behavior in a custom simulator we coded.

III. APPROACH

Queues can be described with Kendall's notation, with four parameters describing the packet arrival distribution, the packet departure distribution, the number of servers / queues, and the queue size / buffer. In our theoretical exposition, we focus on three building-block queues that frequently appear in packet network reasoning:

- **M/M/1:** Poisson arrivals, exponential service, one server, infinite buffer.
- **M/M/1/N:** Poisson arrivals, exponential service, one server, *finite* capacity N (including the one in service).
- **D/D/1:** Deterministic arrivals, deterministic service, one server, infinite buffer.

We first derive and summarize steady-state metrics for these models, then build a toy simulator and simulate a small multi-hop routed testing where routing/rate decisions are informed by the estimated queueing delay.

A. Queueing-theoretic derivations

We use the standard notation: arrival rate λ , service rate μ , and traffic intensity $\rho \triangleq \lambda/\mu$.

1) M/M/1:

2) *M/M/1*: The M/M/1 queue models a single-server queue with an infinite buffer. Furthermore, M/M stands for memoryless (Markov) arrivals and departures, meaning that future arrivals are independent of past arrivals, since the process has no memory of past arrival times.

More concretely, arrivals and departures follow Poisson processes with rates λ and μ , respectively, with the corresponding interarrival and service times being exponentially distributed. Arrivals $X \sim \text{Pois}(\lambda)$ have mean λ and variance λ , with the time between packets following an exponential distribution $\text{Exp}(\lambda)$ with mean $1/\lambda$ and variance $1/\lambda^2$. Similarly, departures follow the same formulas but with rate μ .

The Poisson probability mass function is

$$f(\lambda, x) = \frac{\lambda^x e^{-\lambda}}{x!}. \quad (1)$$

The exponential distribution has probability density function

$$f(\lambda, x) = \lambda e^{-\lambda x}. \quad (2)$$

One important property of Poisson processes leveraged in queueing theory is PASTA (Poisson Arrivals See Time Averages). This states that the expected number of packets seen by an arriving packet is equal to the expected number of packets in the system. This property does not hold for all arrival distributions.

Define

- $p_n(t)$: probability of n packets in the system at time t ,
- p_n : steady-state probability, $p_n = \lim_{t \rightarrow \infty} p_n(t)$.

To derive the steady-state equations, we observe what happens when the number of packets at time t is n , and t is increased by an infinitesimal Δt :

$$\begin{aligned} P(\text{no increase of } n) &= P(0 \text{ arrivals, } 0 \text{ departures}) \\ &\quad + P(1 \text{ arrival, } 1 \text{ departure}) \\ &\quad + P(k \text{ arrivals, } k \text{ departures}) \\ &= (1 - \lambda\Delta t)(1 - \mu\Delta t) + \lambda\Delta t \mu\Delta t \\ &\quad + o(\Delta t) \\ &= (1 - (\lambda + \mu)\Delta t) + o(\Delta t). \end{aligned}$$

$$\begin{aligned} P(\text{decrease of } n \text{ by } 1) &= P(0 \text{ arrivals, } 1 \text{ departure}) \\ &\quad + P(k - 1 \text{ arrivals, } k \text{ departures}) \\ &= \mu\Delta t(1 - \lambda\Delta t) + o(\Delta t) \\ &= \mu\Delta t + o(\Delta t). \end{aligned}$$

Other cases have an infinitesimal ($o(\Delta t)$) probability. Therefore:

$$\begin{aligned} p_n(t + \Delta t) &= p_n(t) \cdot P(\text{no increase of } n) \\ &\quad + p_{n+1}(t) \cdot P(\text{increase of } n \text{ by } 1) \\ &\quad + p_{n-1}(t) \cdot P(\text{decrease of } n \text{ by } 1) \\ &= p_n(t)(1 - (\lambda + \mu)\Delta t) \\ &\quad + p_{n+1}(t)\lambda\Delta t + p_{n-1}(t)\mu\Delta t + o(\Delta t). \end{aligned}$$

$$\begin{aligned} \frac{p_n(t + \Delta t) - p_n(t)}{\Delta t} &= -(\lambda + \mu)p_n(t) \\ &\quad + \lambda p_{n-1}(t) + \mu p_{n+1}(t) + o(\Delta t). \end{aligned}$$

As $dp_n(t)/dt = 0$ in the steady-state case and $\lim \Delta t \rightarrow 0$:

$$(\lambda + \mu)p_n = \lambda p_{n-1} + \mu p_{n+1}.$$

Base case ($n = 0$):

$$p_0\lambda = p_1\mu \Rightarrow p_1 = (\lambda/\mu)p_0.$$

The same argument can show that

$$p_2 = (\lambda/\mu)p_1 = (\lambda/\mu)^2 p_0,$$

and therefore

$$p_n = (\lambda/\mu)^n p_0 = \rho^n p_0,$$

where $\rho = \lambda/\mu$.

$$\begin{aligned} \sum_{n=0}^{\infty} p_n &= 1 \Rightarrow \sum_{n=0}^{\infty} \rho^n p_0 = 1. \\ p_0 &= \frac{1}{\sum_{n=0}^{\infty} \rho^n} = 1 - \rho, \quad |\rho| < 1. \end{aligned}$$

$$p_n = \rho^n (1 - \rho).$$

One important observation is that if the arrival rate λ is bigger than the departure rate μ , $p_n(t)$ will tend towards infinity since more packets enter the system than those that leave.

When a packet arrives to the queue, the expected waiting time is:

$$E[T] = E[W_q] + \frac{1}{\mu},$$

where W_q is the time for all packets ahead to be served.

$$E[T] = \frac{E[N_a]}{\mu} + \frac{1}{\mu},$$

where N_a is the number of packets in the system seen by the arriving packet.

From the PASTA property, we know that $E[N_a] = E[N]$, and from Little's law, we know that $E[N] = \lambda E[T]$.

$$E[T] = \rho E[T] + \frac{1}{\mu}$$

$$E[T] = \frac{1}{\mu(1 - \rho)} = \frac{1}{\mu - \lambda}, \quad \text{where } \mu > \lambda.$$

$$\begin{aligned} E[N] &= \sum_{n=0}^{\infty} n p_n \\ &= (1 - \rho) \sum_{n=1}^{\infty} n \rho^n \\ &= \frac{\rho(1 - \rho)}{(1 - \rho)^2} = \frac{\rho}{1 - \rho} \end{aligned}$$

3) *Finite-buffer M/M/1/N*: Same as the M/M/1 queue, but with a finite buffer of size N instead of an infinite buffer.

$$p_n = \rho^n p_0 \text{ as before, but } 0 \leq n \leq N.$$

$$p_0 = \frac{1}{\sum_{n=0}^N \rho^n} = \frac{1 - \rho}{1 - \rho^{(N+1)}}$$

$$p_n = \frac{\rho^n (1 - \rho)}{1 - \rho^{(N+1)}}, \quad 0 \leq n \leq N$$

As before, $\rho > 1$, but here we can calculate what happens for the edge case when $\rho = 1$.

$$p_0 \sum_{n=0}^N \rho^n = 1 \Rightarrow p_0 = \frac{1}{N+1}$$

$$E[T] = \frac{E[N]}{\lambda(1 - p_N)} = \frac{1 - (N+1)\rho^N + N\rho^{N+1}}{\mu(1 - \rho)(1 - \rho^N)}, \quad \rho \neq 1.$$

$$E[T] = \frac{N+1}{2\lambda}, \quad \rho = 1.$$

$$E[N] = \sum_{n=0}^N n p_n = \frac{(1 - \rho) \sum_{n=1}^N n \rho^n}{1 - \rho^{(N+1)}} = \frac{\rho(1 - (N+1)\rho^N + N\rho^{N+1})}{(1 - \rho^{(N+1)})(1 - \rho)}$$

4) *D/D/1*: Single server queue with an infinite buffer. The key difference with the M/M/1 model is that arrivals and departures follow a deterministic, or degenerate rate, instead of a rate with a random probability. A D/D/1 queue has deterministic interarrival time $a = 1/\lambda$ and deterministic service time $s = 1/\mu$. If $a \geq s$ (equivalently $\lambda \leq \mu$), the server completes each job before the next arrival, so steady-state queueing delay is zero:

$$\mathbb{E}[W_q] = 0, \quad \mathbb{E}[T] = s = \frac{1}{\mu}. \quad (3)$$

The mean number in system follows from Little's law:

$$\mathbb{E}[N] = \lambda \mathbb{E}[T] = \frac{\lambda}{\mu} = \rho. \quad (4)$$

If $a < s$ (i.e., $\lambda > \mu$), the queue is unstable and the backlog grows unbounded.

B. Small multi-hop routed testbed approach

To build the network simulation, we use a Python object-oriented discrete-event simulator. The network is modeled as a directed graph with links (servers) of capacity μ_e (packets/s) and per-node FIFO queues. Flows inject packets at sources according to a configurable process (e.g., Poisson at rate λ_f). Each packet carries a destination; routing chooses the next hop at each node.

Queueing-based routing heuristic. At runtime, we estimate a per-link congestion cost using a queueing-inspired

delay proxy. For link e , let $\hat{\lambda}_e$ be the measured recent arrival rate into that link's queue. We define

$$\hat{d}_e \triangleq \frac{1}{\mu_e - \hat{\lambda}_e + \epsilon},$$

with a small $\epsilon > 0$ to avoid numerical blow-ups. Nodes choose next hops by minimizing estimated remaining path cost (sum of $\hat{d}_e$), updating as queues evolve.

Rate allocation. When multiple candidate outgoing links/routes exist, we split traffic among them inversely proportional to predicted congestion:

$$x_{f,e} \propto \frac{1}{\hat{d}_e} \quad \text{subject to} \quad \sum_f x_{f,e} \leq \mu_e, \quad (5)$$

using a simple normalization/clipping step. (This is a lightweight heuristic, not an optimal controller.)

Algorithm 1 Queueing-based routing and rate allocation (per node, per epoch)

- 1: Measure recent $\hat{\lambda}_e$ for each outgoing link e .
 - 2: Compute $\hat{d}_e \leftarrow 1/(\mu_e - \hat{\lambda}_e + \epsilon)$.
 - 3: For each destination d , compute next-hop costs using shortest path on edge weights $\hat{d}_e$.
 - 4: For packets/flows eligible to transmit, split service among candidate links with weights $\propto 1/\hat{d}_e$, then clip to satisfy capacities.
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Baselines. We compare against:

- **Equal rate allocation:** evenly split traffic across available outgoing options.
- **Static shortest-hop routing:** fixed next-hop choice based on hop count (no queue awareness).

IV. RESULTS AND DISCUSSION

A. Single-queue validation

In isolated single-queue experiments, the simulator reproduces the expected trends:

- For M/M/1 with $\rho < 1$, measured $\mathbb{E}[T]$ increases sharply as $\rho \rightarrow 1$, matching $\mathbb{E}[T] = 1/(\mu - \lambda)$.

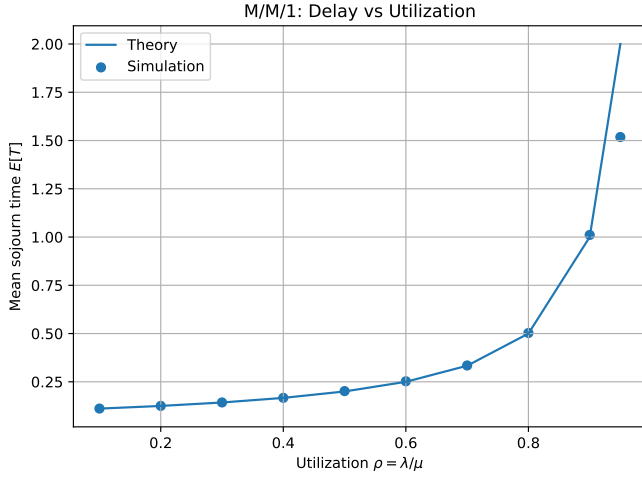


Fig. 1. M/M/1 mean sojourn time versus utilization.

- For M/M/1/N, finite buffering caps delay at the expense of nonzero p_N ; as offered load increases beyond μ , throughput saturates near μ while drop probability rises.

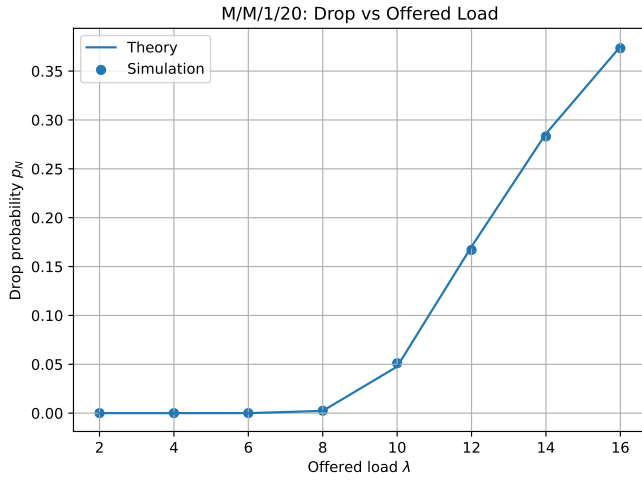


Fig. 2. M/M/1/N drop probability versus offered load.

- For D/D/1 with $\lambda \leq \mu$, queueing delay is approximately zero and packets experience near-constant service time.

B. Multi-hop network evaluation

On small routed topologies with competing flows, queueing-informed decisions reduce the chance that one link becomes a persistent bottleneck. Compared to equal splitting and static routing, the queueing-based heuristic:

- decreases end-to-end delay at moderate-to-high offered load by diverting traffic from congested links,
- reduces drop probability when buffers are finite by preventing prolonged queue build-up at a single node,
- maintains comparable throughput, with improvements most noticeable when multiple feasible paths exist.

Table I summarizes the aggregate network metrics for static shortest-hop routing versus the queue-aware policy.

TABLE I
SMALL-NETWORK SIMULATION COMPARISON ($t_{\text{end}} = 3000\text{s}$,
WARMUP=300S, SEED=1).

Policy	Thr. (pkt/s)	Drop	Delay (s)
Static SH	17.10	0.287	1.504
Queue-aware	19.64	0.183	1.343

Fig. 4 shows mean delay versus offered load for both policies.

Fig. 3 shows aggregate throughput versus offered load.

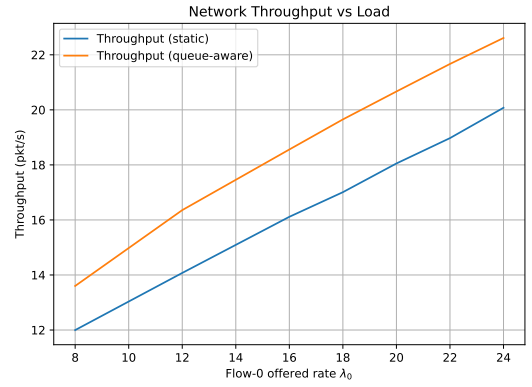


Fig. 3. Aggregate throughput versus offered load (λ_0).

Fig. 4 shows mean delay of delivered packets versus offered load.

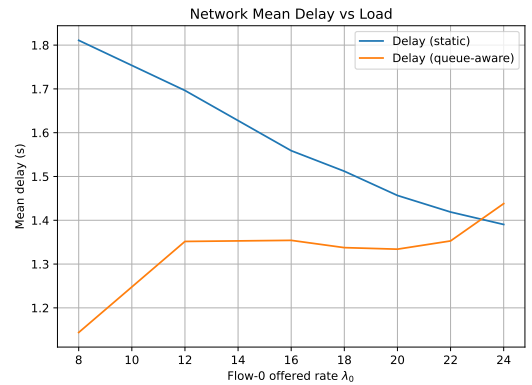


Fig. 4. Mean delay (delivered packets) versus offered load (λ_0).

Fig. 5 shows the drop probability versus offered load.

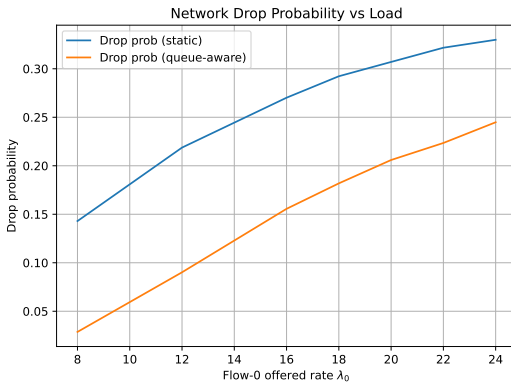


Fig. 5. Drop probability versus offered load (λ_0).

a) *Link-level bottleneck analysis:* Fig. 6 reports the average occupancy on the bottleneck link $1 \rightarrow 4$ versus offered load.

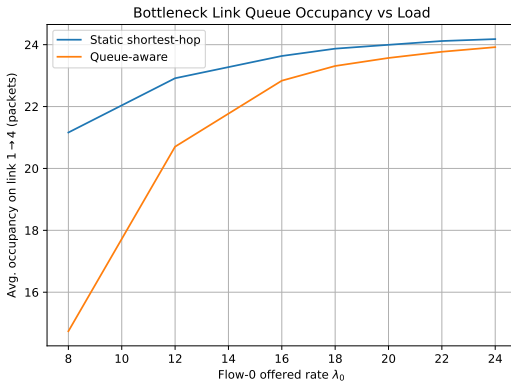


Fig. 6. Average queue occupancy on bottleneck link $1 \rightarrow 4$ versus offered load.

b) *Per-flow performance:* Fig. 7 shows per-flow throughput versus offered load.

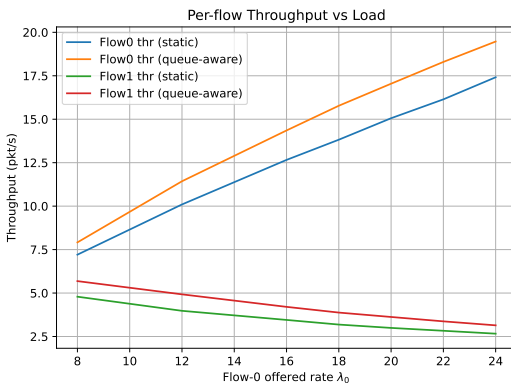


Fig. 7. Per-flow throughput versus offered load.

V. CONCLUSION AND FUTURE WORK

This paper revisited three foundational queueing models (M/M/1, M/M/1/N, and D/D/1) and used their steady-state

insights to guide a practical, queue-aware routing and rate allocation heuristic for small networks. Our discrete-event simulator validates the analytical single-queue behavior and shows that queue-informed control can improve end-to-end delay and loss relative to naive baselines when multiple routing/rate choices exist.

Future work includes (i) extending the heuristic to multi-commodity queue models with per-destination queues, (ii) integrating explicit control objectives (e.g., proportional fairness or delay minimization) with stability guarantees, and (iii) evaluating on ns-3 or emulation testbeds to better match real protocol behaviors and wireless link dynamics.

TEAM PARTICIPATION

Pérez Calderón: abstract, introduction, related work, and overall write-up organization; initial simulator scaffolding.

Aravind: queueing-theoretic derivations cleanup, discrete-event simulator implementation, and results/discussion drafting.

Shrey Parikh: execution and validation of simulation experiments; single-queue (M/M/1, M/M/1/N, D/D/1) theory–simulation verification; comparison of static versus queue-aware routing policies; team poster

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